Sahaj Bhadja

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Summary

Software engineer with 2+ years applying **C/C++, Python, Linux, operating-system fundamentals, multithreading, and distributed systems** to requirements-driven development, defect resolution, testing, integration, and reliable software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Developed fault-tolerant Python services with explicit job-state transitions and failure isolation, processing 1,000+ concurrent tasks while improving throughput 3x and reliably preventing cascading defects across distributed workflows.
- Implemented requirements-driven validation, reconciliation, and error handling across 5,000+ monthly integration jobs, documenting failures and supporting CI/CD releases that sustained 99%+ reliability in production environments at scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Built multithreaded Python processing using ThreadPoolExecutor and secure file ingestion, handling 1,000+ files per batch with fault-tolerant execution and stable performance across 10,000+ daily requests in production.
- Implemented features and resolved defects across React and Flask components, adding automated tests, fallback logic, and diagnostic logging while consistently reducing feedback turnaround 60% for 150+ instructors.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Designed and maintained model-development software pipelines processing 100,000+ autonomous-driving trajectory logs, improving long-horizon planning 18% through established architecture, simulation testing, algorithm evaluation, and iterative integration at scale.
- Integrated spatial-temporal representations across ego, road, and nearby-agent states, improving closed-loop stability 15% while debugging system behavior and documenting reproducible evaluations for senior research collaborators at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed and integrated cross-platform software across Linux, Windows, and macOS, automating executable builds and 20+ unit and integration tests while reliably increasing nightly delivery pipeline efficiency 68%.
- Resolved defects in Java and Python API components using object-oriented design patterns, improving response times 30%, reducing payload sizes 25%, and lowering technical debt 20% through maintainable solutions.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*

- Designed a distributed blockchain algorithm integrating five lattice-based cryptographic primitives, reducing centralized-control and quantum-risk exposure 25% through benchmark simulations, architecture analysis, and iterative software design at scale.
- Evaluated consensus mechanisms against defined security requirements and documented design tradeoffs in a peer-reviewed technical paper, presenting implementation findings and measured results to engineering researchers at IISc Bengaluru.

Skills

Programming: C, C++, Python, Java, JavaScript, TypeScript, SQL, Shell, Bash
Systems: Linux, Unix, operating systems, multithreading, concurrency, distributed systems, scheduling
Design: Object-oriented programming, data structures, algorithms, software architecture, design patterns
Quality: Requirements analysis, defect resolution, code review, unit testing, integration testing, debugging
Delivery: Git, GitHub, CI/CD, Jenkins, Docker, Agile/Scrum, technical documentation, performance analysis

Projects

CPU-scheduling Simulator Live | *C++, Operating Systems*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling in C++, testing 1,000+ process combinations and reducing average turnaround time 25% through algorithm and edge-case optimization.

Chat2SQL Code | *Linux, Software Integration*

- Integrated Java backend and TypeScript frontend modules supporting 10+ databases, reliably configuring Docker, Maven, and Node.js workflows for cross-platform development and debugging across Windows, macOS, and Linux.

FinNexus Live | *Distributed Systems, Real-Time Software*

- Built a distributed TypeScript system supporting 50,000+ users and one million transactions with WebSocket updates, secure APIs, Kubernetes deployment, and reliable, scalable MongoDB access under high-volume workloads.